

Scalability & Future Plan

Date	06 July 2026
Team ID	
Project Name	Ledgerline – Credit Card Approval Prediction System
Maximum Marks	1 Mark

Current System Limitations

S.No	Limitation	Impact	Priority to Address
1	Trained on synthetic data, not real-world applicant records	Model accuracy in production depends on real data	High
2	IBM Watson Machine Learning deployment not yet live	Cloud-hosted scaling not available until real credentials are added	Medium
3	No user authentication or role-based access control	All users see the same UI; no per-role restrictions	Medium
4	No concurrent load-testing performed	Multi-user performance under real traffic is unknown	Medium

Scalability Plan

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-user local Flask dev server	Deploy behind Gunicorn/uWSGI + cloud hosting (IBM Cloud); IBM Watson ML endpoint for stateless horizontal scaling
2	Data Storage	Flat CSV files	Migrate to PostgreSQL or a cloud data warehouse for production; IBM Cloud Object Storage for large batch files
3	Performance	< 1 second per prediction locally (not load-tested)	Load-test with Locust or JMeter; add Redis caching for repeated lookups if needed
4	Security	No authentication	Add JWT-based authentication; role-based access (analyst / compliance / customer); HTTPS via cloud provider certificate

Future Roadmap

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Retrain on the real Kaggle dataset or the bank's historical data	1–2 months post-submission	Significantly improved real-world model accuracy
Phase 3	Complete IBM Watson Machine Learning live deployment with authentication	2–3 months post-submission	Cloud-hosted, access-controlled scoring for real bank use
Phase 4	Add SHAP-based per-applicant explainability (which features drove the decision for each applicant)	3–6 months post-submission	Full explainability for compliance and fair-lending review